

Full Stack Web Development



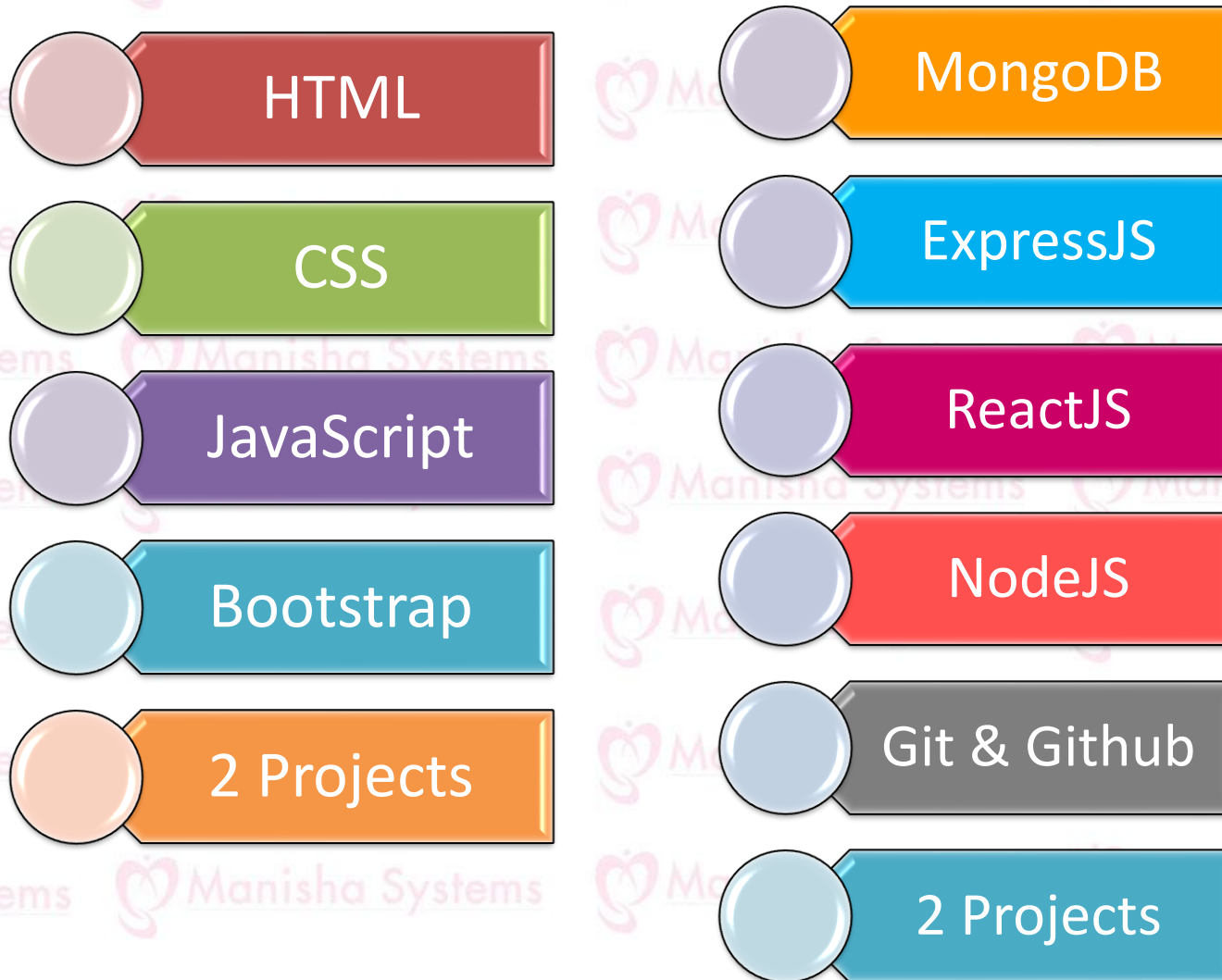
Manisha Systems

CAREER DESIGN MANTRA...

Manisha Systems
Designing Mantra

Full Stack Web Development

Duration : 6 Months





Manisha Systems

Career Designing Mantra

HTML

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CSS

HTML Introduction

Hypertext Markup Language (HTML) is the foundation of website and web application development. HTML is a **computer language** used to allow **website creation** i.e. HTML is a language for describing web pages. It is very **easy to learn**.

- HTML stands for **Hyper Text Markup Language**
- HTML file is a text file containing **mark-up tags**
- The mark-up tags tell the Web browser **how to display** the page
- HTML file must have an **htm** or **html** file extension
- HTML file can be created using a **simple text editor**
- HTML documents are also called **web pages**

How does it work?

HTML consists of a series of short **codes** typed into a text, these are the tags. The text is then **saved as a html file**, and **viewed through a browser**, like **Internet Explorer or Mozilla Firefox**

Programs required to start with HTML :

- A text editor of your choice. If you're using Windows it'll be Notepad or WordPad
- A web browser. A browser is the program that makes it possible to browse and open websites i.e Google Chrome, Microsoft Edge, Mozilla Firefox, Opera etc.



Brave



HTML Tags

HTML markup tags are usually called HTML tags

- HTML tags are keywords (tag names) surrounded by **angle brackets** like <html>
- HTML tags normally **come in pairs** like and
- HTML tags are not case sensitive, means the same as
- The first tag in a pair is the **start tag**, the second tag is the **end tag**
- The end tag is written like the start tag, with a **forward slash** before the tag name
- Start and end tags are also called **opening tags** and **closing tags**

Syntax :

<tagname> content </tagname>

HTML Elements

- "HTML tags" and "HTML elements" are often used to describe the same thing.
- But an HTML element is everything between the start tag and the end tag, including the tags:

HTML Element:

<p> This is a paragraph. </p>

HTML Page Structure

Visualization of an HTML page structure:

```
<html>
```

```
<head>
```

```
<title>Page title</title>
```

```
</head>
```

```
<body>
```

```
<h1>This is a heading</h1>
```

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

```
</body>
```

```
</html>
```


The <!DOCTYPE> Declaration

- The <!DOCTYPE> declaration helps the browser to display a web page correctly.
- There are many different documents on the web, and a browser can only display an HTML page 100% correctly if it knows the HTML type and version used.

Common Declarations HTML5

<!DOCTYPE html>

Doctype

- Explains what type of document the page is
- HTML4, HTML5, XHTML, etc

HTML5

<!DOCTYPE html>

HTML4.01 Strict

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">

XHTML 1.0 Strict

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">

HTML Editors

HTML can be edited by using a professional HTML editor like:

- Sublime Text
- Atom.io
- Visual Studio Code (VS Code)
- Brackets
- NodePad ++ (Windows)
- TextMate (Mac)

However, for learning HTML we prefer a simple text editor like **Notepad** initially and later will switch to VS Code. Following 4 steps will create your first web page with Notepad.

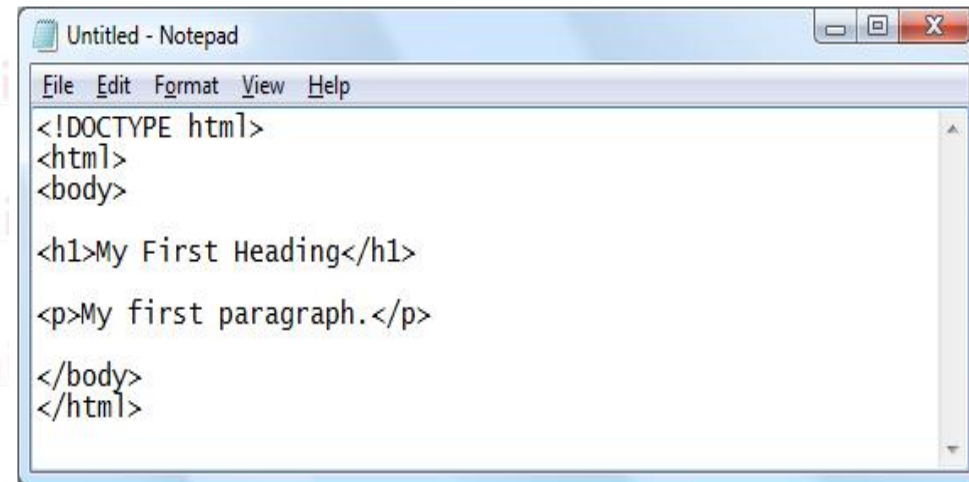
Step 1: Start Notepad

Step 2: Edit Your HTML with Notepad

Type your HTML code into your Notepad:

Step 3: Save Your HTML

Step 4: Run the HTML in Your Browser



- Ctrl+u to see the html code in browser.
- Index.html is the root/home page of a website.
- Hit F12 Function key in Developer Tool to see the html code and css style

Inline Vs Block Level Elements:

Inline Elements:

- Do not start on a new line
- Take only the necessary width

Block Elements:

- Starts on a new line
- Take full width available

Inline Level: <a>, ,

Block Level: <h1>--<h6>, <p>, <div>, <form>

Basic HTML Tags :

Tag	Description
<html>	Defines an HTML document
<body>	Defines the document's body
<h1> to <h6>	Defines header 1 to header 6
<p>	Defines a paragraph

	Inserts a single line break
<hr>	Defines a horizontal rule
<!-->	Defines a comment

HTML Headings

Headings are defined with the <h1> to <h6> tags. <h1> defines the largest heading. <h6> defines the smallest heading. HTML automatically adds an extra blank line before and after a heading.

<h1>This is a heading1</h1>

<h2>This is a heading2</h2>

<h3>This is a heading3</h3>

<h4>This is a heading4</h4>

<h5>This is a heading5</h5>

<h6>This is a heading6</h6>

HTML Paragraphs

HTML paragraphs are defined with the <p> tag. HTML automatically adds an extra blank line before and after a paragraph.

Example

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

Line Breaks

The Line Break (
) tag is used when you want to end a line, but don't want to start a new paragraph. The
 tag forces a line break wherever you place it.

Example

```
<p>This <br> is a para<br>graph with line breaks</p>
```

The
 tag is an empty tag. It has no closing tag.

Comments in HTML

The comment tag is used to insert a comment in the HTML source code. The browser will ignore a comment. You can use comments to explain your code, which can help you when you edit the source code at a later date.

```
<!-- This is a comment -->
```

Note that you need an exclamation point after the opening bracket, but not before the closing bracket.

Example

```
<!-- My comments will be inserted here-->
```


HTML Lines

The <hr> tag creates a horizontal line in an HTML page.

The hr element can be used to separate content:

Example

```
<p>This is a paragraph.</p>
```

```
<hr>
```

```
<p>This is a paragraph.</p>
```

```
<hr>
```

```
<p>This is a paragraph.</p>
```

HTML Output - Useful Tips

- You cannot be sure how HTML will be displayed. Large or small screens, and resized windows will create different results.
- With HTML, you cannot change the output by adding extra spaces or extra lines in your HTML code.
- The browser will remove extra spaces and extra lines when the page is displayed. Any number of lines count as one line, and any number of spaces count as one space.

HTML Attributes

- HTML elements can have **attributes**
- Attributes provide **additional information** about an element
- Attributes are always specified in **the start tag**
- Attributes come in name/value pairs like: **name="value"**

HTML Text Formatting

HTML Formatting Tags

HTML uses tags like `<b>` and `<i>` for formatting output, like **bold** or *italic* text. These HTML tags are called formatting tags

HTML Text Formatting Tags

Tag	Description
<code></code>	Defines bold text
<code></code>	Defines emphasized text
<code><i></code>	Defines text in italics
<code><small></code>	Defines smaller text
<code></code>	Defines important text
<code><sub></code>	Defines subscripted text
<code><sup></code>	Defines superscripted text
<code><ins></code>	Defines inserted text
<code></code>	Defines deleted text
<code><mark></code>	Defines marked/highlighted text

Example :

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<p><b>This text is bold</b></p>
```

```
<p><strong>This text is strong</strong></p>
```

```
<p><em>This text is emphasized</em></p>
```

```
<p><i>This text is italic</i></p>
```

```
<p><small>This text is small</small></p>
```

```
<p>This is<sub> subscript</sub> and <sup>superscript</sup></p>
```

```
</body>
```

```
</html>
```

Output :

This text is bold

This text is strong

This text is emphasized

This text is italic

This text is small

This is subscript and superscript

HTML Links

Links are found in nearly all Web pages. Links allow users to click their way from page to page.

HTML Hyperlinks (Links)

- The HTML <a> tag defines a hyperlink.
- A hyperlink (or link) is a word, group of words, or image that you can click on to jump to another document.
- When you move the cursor over a link in a Web page, the arrow will turn into a little hand.
- The most important attribute of the <a> element is the href attribute, which indicates the link's destination.

By default, links will appear as follows in all browsers:

- An unvisited link is underlined and blue
- A visited link is underlined and purple
- An active link is underlined and red

HTML Link Syntax

The HTML code for a link is simple. It looks like this:

```
<a href="url">Link text</a>
```

The href attribute specifies the destination of a link.

Example

```
<a href="http://www.mkcl.org/">Visit MKCL</a>
```

which will display like this: Visit MKCL

Clicking on this hyperlink will send the user to MKCLs' homepage.

Tip: The "*Link text*" doesn't have to be text. It can be an image or any other HTML element.

```

<!DOCTYPE html>
<html>
<body>

<p>
  <a href="default.asp">HTML Tutorial</a>
  This is a link to a page on this website.
</p>

<p>
  <a href="http://www.mkcl.org/">MKCL</a>
  This is a link to a website on the World Wide
  Web.
</p>

</body>
</html>

```

Output:

[HTML Tutorial](#) This is a link to a page on this website.
[MKCL](#) This is a link to a website on the World Wide
 Web.

HTML Links - The target Attribute

The target attribute specifies where to open the linked document.

The example below will open the linked document in a new browser window or a new tab:

Example

```

<a href="http://www.mkcl.org/" target="_blank">
Visit MKCL!</a>

```

Program

```

<!DOCTYPE html>
<html>
<body>

<a href="http://www. mkcl.org" target="_blank">
Visit MKCL.org!</a>

<p>If you set the target attribute to "_blank", the link will open
in a new browser window/tab.</p>

</body>
</html>

```

Output :

Visit MKCL.org!
 If you set the target attribute to "_blank", the link will open in a
 new browser window/tab.

HTML Links - The id Attribute

The id attribute can be used to create a bookmark inside an HTML document.

Tip: Bookmarks are not displayed in any special way. They are invisible to the reader.

Example

An anchor with an id inside an HTML document:

```
<a id="tips">Useful Tips Section</a>
```

Create a link to the "Useful Tips Section" inside the same document:

```
<a href="#tips">Visit the Useful Tips Section</a>
```

Or, create a link to the "Useful Tips Section" from another page:

```
<a href="http://www.mkcl.org/html_links.htm#tips"> Visit the Useful Tips Section</a>
```


HTML <head>

The HTML <head> Element

- The <head> element is a container for all the head elements. Elements inside <head> can include scripts, instruct the browser where to find style sheets, provide meta information, and more.
- The following tags can be added to the head section: <title>, <style>, <meta>, <link>, <script>, <noscript>, and <base>.

The HTML <title> Element

The <title> tag defines the title of the document.

The <title> element is required in all HTML/XHTML documents.

The <title> element:

- defines a title in the browser toolbar
- provides a title for the page when it is added to favorites
- displays a title for the page in search-engine results

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Title of the document</title>
```

```
</head>
```

```
<body>
```

```
The content of the document.....
```

```
</body>
```

```
</html>
```

HTML Tables

- Tables are defined with the <table> tag.
- A table is divided into rows (with the <tr> tag), and each row is divided into data cells (with the <td> tag). td stands for "table data," and holds the content of a data cell. A <td> tag can contain text, links, images, lists, forms, other tables, etc.

Table Example

```
<table border="1">
```

```
<tr>
```

```
<td>row 1, cell 1</td>
```

```
<td>row 1, cell 2</td>
```

```
</tr>
```

```
<tr>
```

```
<td>row 2, cell 1</td>
```

```
<td>row 2, cell 2</td>
```

```
</tr>
```

```
</table>
```

row 1, cell 1	row 1, cell 2
row 2, cell 1	row 2, cell 2

HTML Tables and the Border Attribute

If you do not specify a border attribute, the table will be displayed without borders. Sometimes this can be useful, but most of the time, we want the borders to show.

To display a table with borders, specify the border attribute:

```
<table border="1">
<tr>
<td>Row 1, cell 1</td>
<td>Row 1, cell 2</td>
</tr>
</table>
```

HTML Table Headers

Header information in a table are defined with the <th> tag.

All major browsers display the text in the <th> element as bold and centered.

```
<table border="1">
<tr>
<th>Header 1</th>
<th>Header 2</th>
</tr>
<tr>
<td>row 1, cell 1</td>
<td>row 1, cell 2</td>
</tr>
<tr>
<td>row 2, cell 1</td>
<td>row 2, cell 2</td>
</tr>
</table>
```

Header 1	Header 2
row 1, cell 1	row 1, cell 2
row 2, cell 1	row 2, cell 2

Empty Cells in a Table

Table cells with no content are not displayed very well in most browsers.

```
<table border="1">
<tr>
<td>row 1, cell 1</td>
<td>row 1, cell 2</td>
</tr>
<tr>
<td>row 2, cell 1</td>
<td></td>
</tr>
</table>
```

row 1, cell 1

row 1, cell 2

row 2, cell 1

Note that the borders around the empty table cell are missing (NB! Mozilla Firefox displays the border).

To avoid this, add a non-breaking space () to empty data cells, to make the borders visible:

```
<table border="1">
<tr>
<td>row 1, cell 1</td>
<td>row 1, cell 2</td>
</tr>
<tr>
<td>row 2, cell 1</td>
<td>&nbsp;</td>
</tr>
</table>
```

row 1, cell 1	row 1, cell 2
row 2, cell 1	

Table With Caption

```
<!DOCTYPE html>
<html>
<body>
<table border="1">
  <caption>Monthly savings</caption>
  <tr>
    <th>Month</th>
    <th>Savings</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
  </tr>
  <tr>
    <td>February</td>
    <td>$50</td>
  </tr>
</table>
</body>
</html>
```

Monthly savings

Month	Savings
January	\$100
February	\$50

Table cells that span more than one row or one column

```
<!DOCTYPE html>
<html>
<body>
<h4>Cell that spans two columns:</h4>
<table border="1">
  <tr>
    <th>Name</th>
    <th colspan="2">Telephone</th>
  </tr>
  <tr>
    <td>Shivani</td>
    <td>55577854</td>
    <td>55577855</td>
  </tr>
</table>

<h4>Cell that spans two rows:</h4>
<table border="1">
  <tr>
    <th>First Name:</th>
    <td>Shivani</td>
  </tr>
  <tr>
    <th rowspan="2">Telephone:</th>
    <td>55577854</td>
  </tr>
  <tr>
    <td>55577855</td>
  </tr>
</table>
</body>
</html>
```

Cell that spans two columns:

Name	Telephone	
Shivani	55577854	55577855

Cell that spans two rows:

First Name:	Shivani
Telephone:	55577854
	55577855

To Add a background to a table.

```
<!DOCTYPE html>
<html>
<body>
```

```
<h4>A background color:</h4>
<table border="1" bgcolor="red">
<tr>
  <td>First</td>
  <td>Row</td>
</tr>
<tr>
  <td>Second</td>
  <td>Row</td>
</tr>
</table>
</body>
</html>
```

A background color:

First	Row
Second	Row

to add a background to one or more table cells

```
<!DOCTYPE html>
<html>
<body>
```

```
<h4>Background color for a cell:</h4>
<table border="1">
<tr>
  <td bgcolor="red">First</td>
  <td>Row</td>
</tr>
<tr>
  <td>Second</td>
  <td>Row</td>
</tr>
</table>
</body>
</html>
```

Background color for a cell:

First	Row
Second	Row

To display elements inside other elements

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<table border="1">
```

```
<tr>
```

```
<td>
```

```
<p>This is a paragraph</p>
```

```
<p>This is another paragraph</p>
```

```
</td>
```

```
<td>This cell contains a table:
```

```
<table border="1">
```

```
<tr>
```

```
<td>A</td>
```

```
<td>B</td>
```

```
</tr>
```

```
<tr>
```

```
<td>C</td>
```

```
<td>D</td>
```

```
</tr>
```

```
</table>
```

```
</td>
```

```
</tr>
```

```
<tr>
```

```
<td>This cell contains a list
```

```
<ul>
```

```
<li>apples</li>
```

```
<li>bananas</li>
```

```
<li>pineapples</li>
```

```
</ul>
```

```
</td>
```

```
<td>HELLO</td>
```

```
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

This is a paragraph This is another paragraph	This cell contains a table: <table><tr><td>A</td><td>B</td></tr><tr><td>C</td><td>D</td></tr></table>	A	B	C	D
A	B				
C	D				
This cell contains a list - apples - bananas - pineapples	HELLO				

```

<!DOCTYPE html>
<html>
<body>

<table width="400" border="1">
<tr>
<th align="left">Money spent on....</th>
<th align="right">January</th>
<th align="right">February</th>
</tr>
<tr>
<td align="left">Clothes</td>
<td align="right">$241.10</td>
<td align="right">$50.20</td>
</tr>
<tr>
<td align="left">Make-Up</td>
<td align="right">$30.00</td>
<td align="right">$44.45</td>
</tr>
<tr>
<td align="left">Food</td>
<td align="right">$730.40</td>
<td align="right">$650.00</td>
</tr>

```

```

<tr>
<th align="left">Sum</th>
<th align="right">$1001.50</th>
<th align="right">$744.65</th>
</tr>
</table>

</body>
</html>

```

Money spent on....	January	February
Clothes	\$241.10	\$50.20
Make-Up	\$30.00	\$44.45
Food	\$730.40	\$650.00
Sum	\$1001.50	\$744.65

Tag	Description
<u><table></u>	Defines a table
<u><th></u>	Defines a table header
<u><tr></u>	Defines a table row
<u><td></u>	Defines a table cell
<u><caption></u>	Defines a table caption
<u><colgroup></u>	Defines groups of table columns
<u><col></u>	Defines the attribute values for one or more columns in a table
<u><thead></u>	Defines a table head
<u><tbody></u>	Defines a table body
<u><tfoot></u>	Defines a table footer

HTML Lists

HTML supports ordered, unordered and definition lists.

Unordered Lists

An unordered list is a list of items. The list items are marked with bullets (typically small black circles).

An unordered list starts with the **** tag.

Each list item starts with the **** tag.

Inside a list item you can put paragraphs, line breaks, images, links, other lists, etc.

```
<ul>
<li>Shivani</li>
<li>Gaurav</li>
<li>Asmita</li>
</ul>
```

- Shivani
- Gaurav
- Asmita

Ordered Lists

An ordered list is also a list of items. The list items are marked with numbers.

An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag.

Inside a list item you can put paragraphs, line breaks, images, links, other lists, etc.

```
<ol>
  <li>Shivani</li>
  <li>Gaurav</li>
  <li>Asmita</li>
</ol>
```

1. Shivani
2. Gaurav
3. Asmita

Definition Lists

- A definition list is not a list of single items. It is a list of items (terms), with a description of each item (term).
- A definition list starts with a `<dl>` tag (**definition list**).
- Each term starts with a `<dt>` tag (**definition term**).
- Each description starts with a `<dd>` tag (**definition description**).
- Inside the `<dd>` tag you can put paragraphs, line breaks, images, links, other lists, etc.

```
<dl>
  <dt>Shivani</dt>
  <dd> HSC Student</dd>
  <dt>Gaurav</dt>
  <dd>B Tech Student</dd>
</dl>
```

Shivani

HSC Student

Gaurav

B Tech Student

```
<!DOCTYPE html>
<html>
<body>
```

```
<h4>Disc bullets list:</h4>
<ul style="list-style-type:disc">
  <li>Apples</li>
  <li>Bananas</li>
  <li>Lemons</li>
  <li>Oranges</li>
</ul>
```

```
<h4>Circle bullets list:</h4>
<ul style="list-style-type:circle">
  <li>Apples</li>
  <li>Bananas</li>
  <li>Lemons</li>
  <li>Oranges</li>
</ul>
```

```
<h4>Square bullets list:</h4>
<ul style="list-style-type:square">
  <li>Apples</li>
  <li>Bananas</li>
  <li>Lemons</li>
  <li>Oranges</li>
</ul>
```

```
</body>
</html>
```

Disc bullets list:

- Apples
- Bananas
- Lemons
- Oranges

Circle bullets list:

- Apples
- Bananas
- Lemons
- Oranges

Square bullets list:

- Apples
- Bananas
- Lemons
- Oranges

```
<!DOCTYPE html>
<html>
<body>
```

```
<h4>Numbered list:</h4>
<ol>
<li>Apples</li>
<li>Bananas</li>
<li>Lemons</li>
<li>Oranges</li>
</ol>
```

```
<h4>Letters list:</h4>
<ol type="A">
<li>Apples</li>
<li>Bananas</li>
<li>Lemons</li>
<li>Oranges</li>
</ol>
```

```
<h4>Lowercase letters list:</h4>
<ol type="a">
<li>Apples</li>
<li>Bananas</li>
<li>Lemons</li>
<li>Oranges</li>
</ol>
```

```
<h4>Roman numbers list:</h4>
<ol type="I">
<li>Apples</li>
<li>Bananas</li>
<li>Lemons</li>
<li>Oranges</li>
</ol>
```

```
<h4>Lowercase Roman numbers list:</h4>
<ol type="i">
<li>Apples</li>
<li>Bananas</li>
<li>Lemons</li>
<li>Oranges</li>
</ol>
```

```
</body>
</html>
```

Numbered list:

1. Apples
2. Bananas
3. Lemons
4. Oranges

Letters list:

- A. Apples
- B. Bananas
- C. Lemons
- D. Oranges

Lowercase letters list:

- a. Apples
- b. Bananas
- c. Lemons
- d. Oranges

Roman numbers list:

- I. Apples
- II. Bananas
- III. Lemons
- IV. Oranges

Lowercase Roman numbers list:

- i. Apples
- ii. Bananas
- iii. Lemons
- iv. Oranges

Nest lists

```
<!DOCTYPE html>
<html>
<body>

<h4>A nested List:</h4>
<ul>
<li>Coffee</li>
<li>Tea
  <ul>
<li>Black tea</li>
<li>Green tea</li>
</ul>
</li>
<li>Milk</li>
</ul>

</body>
</html>
```

A nested List:

- Coffee
- Tea
 - Black tea
 - Green tea
- Milk

```
<!DOCTYPE html>
<html>
<body>

<h4>A nested List:</h4>
<ul>
<li>Coffee</li>
<li>Tea
  <ul>
<li>Black tea</li>
<li>Green tea
    <ul>
<li>China</li>
<li>Africa</li>
</ul>
</li>
</ul>
</li>
<li>Milk</li>
</ul>

</body>
</html>
```

A nested List:

- Coffee
- Tea
 - Black tea
 - Green tea
- China
- Africa
- Milk

A Definition List:

```
<!DOCTYPE html>
<html>
<body>

<h4>A Definition List:</h4>
<dl>
<dt>Coffee</dt>
<dd>- black hot drink</dd>
<dt>Milk</dt>
<dd>- white cold drink</dd>
</dl>

</body>
</html>
```

A Definition List:

- Coffee
- black hot drink
- Milk
- white cold drink

HTML Images

- In HTML, images are defined with the tag.
- The tag is empty, which means that it contains attributes only and it has no closing tag.
- To display an image on a page, you need to use the src attribute. Src stands for "source". The value of the src attribute is the URL of the image you want to display on your page.

The syntax of defining an image:

- The URL points to the location where the image is stored. An image named "boat.gif" located in the directory "images" on "www. XX " has the URL: http://www. XXX /images/boat.gif.
- The browser puts the image where the image tag occurs in the document. If you put an image tag between two paragraphs, the browser shows the first paragraph, then the image, and then the second paragraph.

The Alt Attribute

The alt attribute is used to define an "alternate text" for an image. The value of the alt attribute is an author-defined text:

```

```

The "alt" attribute tells the reader what he or she is missing on a page if the browser can't load images. The browser will then display the alternate text instead of the image. It is a good practice to include the "alt" attribute for each image on a page, to improve the display and usefulness of your document for people who have text-only browsers.

Note : If an HTML file contains ten images - eleven files are required to display the page right. Loading images take time, so best advice is: Use images carefully.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<p>
```

An image:

```
</p>
```

An image:



```
<p>
```

A moving image:

```
</p>
```

A moving image:



```
</body>
```

```
</html>
```



```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Cute Birds....</h2>
```

```

```

```
</body>
```

```
</html>
```

Align an image within the text

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h4>Image with default alignment (align="bottom"):</h4>
```

```
<p>This is some text.  This is some text.</p>
```

```
<h4>Image with align="middle":</h4>
```

```
<p>This is some text.  This is some text.</p>
```

```
<h4>Image with align="top":</h4>
```

```
<p>This is some text.  This is some text.</p>
```

```
<p><b>Note:</b> The align attribute is deprecated in HTML 4, and is not supported in
HTML5. Use CSS instead.</p>
```

```
</body>
```

```
</html>
```

An image float to the left or right of a paragraph

```
<!DOCTYPE html>
<html>
<body>

<p>
 A
paragraph with an image. The image will float to the
left of this text.
</p>

<p>
 A
paragraph with an image. The image will float to the
right of this text.
</p>

<p><b>Note:</b> Here we have used the CSS "float"
property to align the image; as the align attribute is
deprecated in HTML 4, and is not supported in
HTML5.</p>

</body>
</html>
```

Use an image as a link

```
<!DOCTYPE html>
<html>
<body>

<p>Create a link of an image:
<a href="http://www.mkcl.org">
</a></p>

<p>No border around the image, but still a link:
<a href="www.mkcl.org">
</a></p>

</body>
</html>
```

```

<!DOCTYPE html>
<html>
<body>

<p>Click on the sun or on one of the planets to watch it closer:</p>



<map name="planetmap">
  <area shape="rect" coords="0,0,82,126" alt="Sun" href="sun.htm">
  <area shape="circle" coords="90,58,3" alt="Mercury" href="mercur.htm">
  <area shape="circle" coords="124,58,8" alt="Venus" href="venus.htm">
</map>

</body>
</html>

```

Tag	Description
<u></u>	Defines an image
<u><map></u>	Defines an image map
<u><area></u>	Defines a clickable area inside an image map

HTML Styles

The style attribute is a new HTML attribute. It introduces CSS to HTML.

The purpose of the style attribute is:

- To provide a common way to style all HTML elements.
- Styles was introduced with HTML 4, as the new and preferred way to style HTML elements. With HTML styles, styles can be added to HTML elements directly by using the style attribute, or indirectly by in separate style sheets (CSS files).

HTML Style Examples

`style="background-color:yellow"`

`style="font-size:10px"`

`style="font-family:Times"`

`style="text-align:center"`

Deprecated Tags and Attributes

In HTML 4, some tags and attributes are defined as deprecated. Deprecated means that they will not be supported in future versions of HTML and XHTML.

The message is clear: Avoid the use of deprecated tags and attributes.

These tags and attributes should be avoided. Use styles instead

Tags	Description
<center>	Defines centered content
 and <basefont>	Defines HTML fonts
<s> and <strikeout>	Defines strikeout text
<u>	Defines underlined text
Attributes	Description
align	Defines the alignment of text
bgcolor	Defines the background color
color	Defines the text color

HTML – Marquee Tag

An HTML marquee is a scrolling piece of text displayed either horizontally across or vertically down your webpage depending on the settings. This is created by using HTML <marquees> tag.

Note – The <marquee> tag deprecated in HTML5. Do not use this element, instead you can use JavaScript and CSS to create such effects.

Syntax

```
<marquee attribute_name = "attribute_value"....more
attributes> One or more lines or text message or image
</marquee>
```

The different attributes of <marquee> tag are:

Sr.No	Attribute & Description
1	width This specifies the width of the marquee. This can be a value like 10 or 20% etc.
2	height This specifies the height of the marquee. This can be a value like 10 or 20% etc.

Sr.No	Attribute & Description
3	direction This specifies the direction in which marquee should scroll. This can be a value like <i>up</i> , <i>down</i> , <i>left</i> or <i>right</i> .
4	behavior This specifies the type of scrolling of the marquee. This can have a value like <i>scroll</i> , <i>slide</i> and <i>alternate</i> .
5	scrollDelay This specifies how long to delay between each jump. This will have a value like 10 etc.
6	scrollAmount This specifies the speed of marquee text. This can have a value like 10 etc.
7	loop This specifies how many times to loop. The default value is INFINITE, which means that the marquee loops endlessly.
8	bgcolor This specifies background color in terms of color name or color hex value.
9	hspace This specifies horizontal space around the marquee. This can be a value like 10 or 20% etc.
10	vspace This specifies vertical space around the marquee. This can be a value like 10 or 20% etc.

Example :

```
<marquee>Manisha Systems, 54, Bajaj Nagar, Nagpur-10 </marquee>
```

```
<marquee width = "50%">This example will take only 50%  
width</marquee>
```

```
<marquee direction = "right">This text will scroll from left to  
right</marquee>
```

```
<marquee direction = "up">This text will scroll from bottom to  
up</marquee>
```

HTML Forms

Student Registration Form

Roll no. :

Student name : -

Father's name :

Date of birth : - - (DD-MM-YYYY)

Mobile no. : -

Email id :

Password :

Gender : ☐ Male ☐ Female

Department : ☐ CSE ☐ IT ☐ ECE ☐ Civil ☐ Mech

Course :

Student photo : No file chosen

City :

Address :

Step 1: Your details

Name

Email

Phone

Step 2: Delivery address

Address

Post code

Country

Step 3: Card details

Card type
☐  VISA ☐  AmEx ☐  Mastercard

Card number

Security code

Name on card

BUY IT!

FORM ELEMENTS

THERE ARE 3 MAIN HTML ELEMENTS THAT ALLOW USERS TO ENTER OR SELECT DATA IN A FORM

John Smith

INPUTS

Enter text

TEXT AREA

▼
EUR
AUD
GBP
USD

DROPDOWN

HTML Forms are used to select different kinds of user input who visits the website. Forms collect and submit user provided data and send it to a web server for processing.

Forms

- A form is an area that can contain form elements, means Forms act as a container for various form elements that allow users to input different types of data.
- Form elements are elements that allow the user to enter information in a form.
- A form is defined with the **<form>** tag.

Syntax :

```
<form>  
.  
  input elements  
.  
</form>
```

The **<form>** element can contain one or more of the following form elements:

- <input>
- <textarea>
- <button>
- <select>
- <option>
- <optgroup>
- <fieldset>
- <label>
- <output>

Attributes: There are many attributes that are associated with the <form> tag.

Attribute	Value	Description
accept	MIME_type	Specifies a comma-separated list of content types that the server accepts.
accept-charset	charset list	Specifies a list of character encodings that the server accepts. The default value is "unknown".
action	URL	Specifies a URI/URL of the back-end script that will process the form
autocomplete	on off	Specifies whether form should have autocomplete on or off
enctype	mimetypes	The mime type used to encode the content of the form.
method	get post	•Specifies the HTTP method to use when the form is submitted. - Possible values –get (the form data is appended to the URL when submitted) •post (the form data is not appended to the URL)
name	form name	Defines a unique name for the form.
novalidate	novalidate 	Specifies that the form should not be validated when submitted.
target	_blank _self _parent _top	Target to open the given URL. _blank – the target URL will open in a new window. _self – the target URL will open in the same frame as it was clicked. _parent – the target URL will open in the parent frameset. _top – the target URL will open in the full body of the window.

Input Tag

The `<input>` tag specifies an input field where the user can enter data. The type of input is specified with the type attribute. The `<input>` element can be displayed in several ways, depending on the type attribute.

The different input types are as follows:

- `<input type="button">`
- `<input type="checkbox">`
- `<input type="color">`
- `<input type="date">`
- `<input type="datetime-local">`
- `<input type="email">`
- `<input type="file">`
- `<input type="hidden">`
- `<input type="image">`
- `<input type="month">`
- `<input type="number">`
- `<input type="password">`
- `<input type="radio">`
- `<input type="range">`
- `<input type="reset">`
- `<input type="search">`
- `<input type="submit">`
- `<input type="tel">`
- `<input type="text">` (default value)
- `<input type="time">`
- `<input type="url">`
- `<input type="week">`

`<form>`

First name:

`<input type="text" name="firstname">`

`<br>`

Last name:

`<input type="text" name="lastname">`

`</form>`

First name:

Last name:

Sr.No	Attribute & Description
1	type Indicates the type of input control and for text input control it will be set to text .
2	name Used to give a name to the control which is sent to the server to be recognized and get the value.
3	value This can be used to provide an initial value inside the control.
4	size Allows to specify the width of the text-input control in terms of characters.
5	maxlength Allows to specify the maximum number of characters a user can enter into the text box.

Label Tag

The **<label>** HTML element represents a caption for an item in a user interface



In above case if we use for and id attributes like this then label is connected to this input.

Alternatively, you can nest the `<input>` directly inside the `<label>`, in which case the `for` and `id` attributes are not needed because the association is implicit.

```
<label>
```

Do you like Nagpur?

```
<input type="checkbox" name="Nagpur">
```

```
</label>
```


Radio Buttons

Radio Buttons are used when you want the user to select one of a limited number of choices.

```
<form>
<input type="radio" name="Gender" value="male" /> Male
<br>
<input type="radio" name="Gender" value="female" /> Female
</form>
```

☐ Male

☐ Female

Checkboxes

Checkboxes are used when you want the user to select one or more options of a limited number of choices.

```
<form>
I have a bike:
<input type="checkbox" name="vehicle" value="Bike" />
<br>
I have a car:
<input type="checkbox" name="vehicle" value="Car" />
<br>
I have an airplane:
<input type="checkbox" name="vehicle" value="Airplane" />
</form>
```

I have a bike: ☐

I have a car: ☐

I have an airplane: ☐

The Form's Action Attribute and the Submit Button

When the user clicks on the "Submit" button, the content of the form is sent to the server. The form's action attribute defines the name of the file to send the content to. The file defined in the action attribute usually does something with the received input

```
<form name="input" action="html_form_submit.asp"
method="get">
Username:
<input type="text" name="user" />
<input type="submit" value="Submit" />
</form>
```

Username:

If you type some characters in the text field above, and click the "Submit" button, the browser will send your input to a page called "html_form_submit.asp". The page will show you the received input.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<form action="">
```

```
First name: <input type="text" name="firstname"><br>
```

```
Last name: <input type="text" name="lastname">
```

```
</form>
```

First name:
Last name:

Note: The form itself is not visible. Also note that the default width of a text field is 20 characters.

<p>Note: The form itself is not visible. Also note that the default width of a text field is 20 characters.</p>

```
</body>
```

```
</html>
```

What is Dropdown?

It allows the user to choose from predefined options.

THE SELECT AND OPTION TAGS

DROPPDOWN

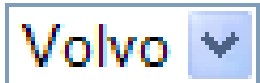
```
<label for="journal-category">Journal Category</label>
<select id="journal-category" name="category" >
  <option value="task">Task</option>
  <option value="note">Note</option>
  <option value="idea">Idea</option>
</select>
```

SETS THE DATA THAT WILL BE
SENT TO THE SERVER

Example : simple drop-down box on an HTML page. A drop-down box is a selectable list.

```
<!DOCTYPE html>
<html>
<body>

<form action="">
<select name="cars">
<option value="Volvo">Volvo</option>
<option value="TATA">TATA</option>
<option value="KIA">KIA</option>
<option value="Audi">Audi</option>
</select>
</form>
</body>
</html>
```



Example : simple drop-down box with a pre-selected value

```
<!DOCTYPE html>
<html>
<body>

<form action="">
<select name="cars">
<option value="Volvo">Volvo</option>
<option value="TATA">TATA</option>
<option value="KIA" selected>KIA</option>
<option value="Audi">Audi</option>
</select>
</form>

</body>
</html>
```



Example : Create a button. On the button you can define your own text

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<form action="">
```

```
<input type="button" value="Hello world!">
```

```
</form>
```

```
</body>
```

```
</html>
```



Example : to draw a border with a caption around your data

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<form action="">
```

```
<fieldset>
```

```
<legend>Personal information:</legend>
```

```
Name: <input type="text" size="30"><br>
```

```
E-mail: <input type="text" size="30"><br>
```

```
Date of birth: <input type="text" size="10">
```

```
</fieldset>
```

```
</form>
```

```
</body>
```

```
</html>
```

Personal information:

Name:

E-mail:

Date of birth:

Example : Input fields and a submit button.

```
<!DOCTYPE html>
<html>
<body>

<form name="input" action="html_form_action.asp"
method="get">
First name: <input type="text" name="FirstName"
value="Mickey"><br>
Last name: <input type="text" name="LastName"
value="Mouse"><br>
<input type="submit" value="Submit">
</form>
```

<p>If you click the "Submit" button, the form-data will be sent to a page called "html_form_action.asp".</p>

```
</body>
</html>
```

First name: Mickey

Last name: Mouse

Submit

Example : Checkboxes, and a submit button

```
<!DOCTYPE html>
<html>
<body>

<form name="input" action="html_form_action.asp"
method="get">
<input type="checkbox" name="vehicle"
value="Bike">I have a bike<br>
<input type="checkbox" name="vehicle" value="Car">I
have a car
<br><br>
<input type="submit" value="Submit">
</form>

<p>If you click the "Submit" button, the form-data will
be sent to a page called "html_form_action.asp".</p>

</body>
</html>
```

☐ I have a bike
☐ I have a car

Submit

Example : radio buttons, and a submit button.

```
<!DOCTYPE html>
<html>
<body>

<form name="input" action="html_form_action.asp"
method="get">
<input type="radio" name="sex" value="male">Male<br>
<input type="radio" name="sex" value="female">Female<br>
<input type="submit" value="Submit">
</form>

<p>If you click the "Submit" button, the form-data will be sent
to a page called "html_form_action.asp".</p>

</body>
</html>
```

☐ Male
☐ Female
Submit

Example : To send e-mail from a form

```
<!DOCTYPE html>
<html>
<body>
<h3>Send e-mail to someone@example.com:</h3>

<form action="MAILTO:someone@example.com"
method="post" enctype="text/plain">
Name:<br>
<input type="text" name="name" value="your
name"><br>
E-mail:<br>
<input type="text" name="mail" value="your
email"><br>
Comment:<br>
<input type="text" name="comment" value="your
comment" size="50"><br><br>
<input type="submit" value="Send">
<input type="reset" value="Reset">
</form>
</body>
</html>
```

Send e-mail to someone@example.com:

Name:

E-mail:

Comment:

Buttons

Buttons trigger action for example submitting a form.

The <button> element in HTML is used to create **clickable buttons**. It is versatile and can be styled and scripted for various functionalities.

Syntax **<button type="Submit">Submit</button>**

Attributes

type: Specifies the button type. Common values:

button: A simple button with no default behavior.

submit: Submits the form data.

reset: Resets the form fields to their initial values.

disabled: Disables the button, making it unclickable.

name and **value**: Used to identify the button when submitting form data.

Buttons Vs. Hyperlinks

Hyperlinks and **Buttons** can appear visually the same but serve different purposes

Hyperlink Element

Get Started

Used to Navigate users to a different page

Often looks like a button element

Button Element

<button type="Submit">Submit</button>

Used to trigger actions like submit form

Can result in navigation to a different page

HTML <textarea> Tag

- The <textarea> tag defines a multi-line text input control.
- The <textarea> element is often used in a form, to collect user inputs like comments or reviews.
- A text area can hold an unlimited number of characters, and the text renders in a fixed-width font (usually Courier).
- The size of a text area is specified by the **cols** and **rows** attributes (or with CSS).
- The **name** attribute is needed to reference the form data after the form is submitted (if you omit the **name** attribute, no data from the text area will be submitted).
- The **id** attribute is needed to associate the text area with a label.

```
<textarea id="journal-entry" name="entry" rows="5" cols="30"></textarea>
```

SETS THE VISIBLE
DIMENSIONS IN TEXT
CHARACTERS

Attribute	Value	Description
autofocus	autofocus	Specifies that a text area should automatically get focus when the page loads
cols	<i>number</i>	Specifies the visible width of a text area
dirname	<i>textarea.name.dir</i>	Specifies that the text direction of the textarea will be submitted
disabled	disabled	Specifies that a text area should be disabled
form	<i>form_id</i>	Specifies which form the text area belongs to
maxlength	<i>number</i>	Specifies the maximum number of characters allowed in the text area
name	<i>text</i>	Specifies a name for a text area
placeholder	<i>text</i>	Specifies a short hint that describes the expected value of a text area
readonly	readonly	Specifies that a text area should be read-only
required	required	Specifies that a text area is required/must be filled out
rows	<i>number</i>	Specifies the visible number of lines in a text area
wrap	hard soft	Specifies how the text in a text area is to be wrapped when submitted in a form

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Textarea tag</title>
```

```
</head>
```

```
<body>
```

```
<h1>HTML Textarea tag </h1>
```

```
<form action="#">
```

```
<textarea rows="8" cols="50" name="blog"></textarea>
```

```
<br>
```

```
<input type="submit" value="submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

HTML Textarea tag

```
<form action="/api/v1/journals" method="post"> ← FORM ELEMENT
```

```
<label for="main-title">Journal Title</label> ← LABEL ELEMENT
```

```
<input id="main-title" type="text" name="title" placeholder="Enter title"> ← INPUT ELEMENT
```

```
<label for="journal-entry">Journal Entry</label> ← LABEL ELEMENT
```

```
<textarea id="journal-entry" name="entry" rows="5" cols="30"></textarea> ← TEXTAREA ELEMENT
```

```
</form>
```



```
<form action="/api/v1/journals" method="post"> ← FORM ELEMENT

<label for="main-title">Journal Title</label> ← LABEL ELEMENT
<input id="main-title" type="text" name="title" placeholder="Enter title"> ← INPUT ELEMENT

<label for="journal-category">Journal Category</label> ← LABEL ELEMENT
<select id="journal-category" name="category"> ← SELECTELEMENT
  <option value="task">Task</option> ← OPTION ELEMENT
  <option value="note">Note</option> ← OPTION ELEMENT
  <option value="idea">Idea</option> ← OPTION ELEMENT
</select>

<label for="journal-entry">Journal Entry</label> ← LABEL ELEMENT
<textarea id="journal-entry" name="entry" rows="5" cols="30"></textarea> ← TEXTAREA ELEMENT

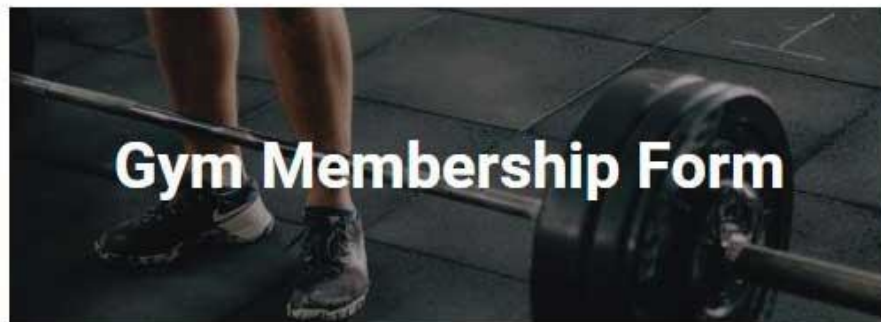
</form>
```

Login Form

Username

Password

Login



Gym Membership Form

Want to become a member of our Gym? Then start by filling our form to complete registration. We will contact you shortly to notify you about your membership card.

Name*

First

Last

Gender*

☐ Male ☐ Female

Your current weight (lbs)*

Desired weight (lbs)*

Height*

BMI*

Home Address*

Street address

City

Region

Postal / Zip code

Country

Email*

Phone

Do you require a personal trainer?*

☐ Yes ☐ No

Have you been in a Gym before?*

☐ Yes ☐ No

Membership Type*

☐ Regular ☐ Pro ☐ VIP

Apply

HTML <div> Tag

- The `<div>` tag defines a division or a section in an HTML document.
- The `<div>` tag is used as a container for HTML elements - which is then styled with CSS or manipulated with JavaScript.
- The `<div>` tag is easily styled by using the class or id attribute.
- Any sort of content can be put inside the `<div>` tag!
- By default, browsers always place a line break before and after the `<div>` element.

`<div>`

`</div>`

Example:

```
<html>
<head>
<style>
.myDiv {
    border: 5px outset red;
    background-color: lightblue;
    text-align: center;
}
</style>
</head>
<body>

<div class="myDiv">
    <h2>This is a heading in a div element
</h2>
    <p>This is some text in a div element.</p>
</div>

</body>
</html>
```

HTML5 Semantic Tags

A semantic element clearly describes its meaning to both the browser and the developer

`<header></header>`

`<footer></footer>`

`<aside></aside>`

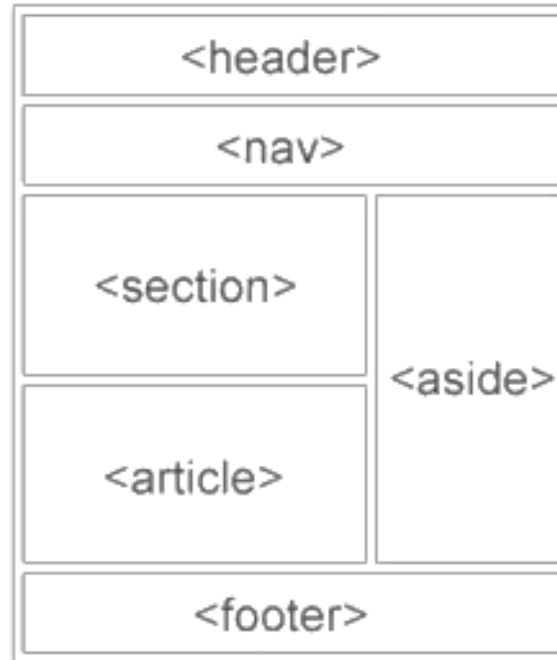
`<main></main>`

`<article></article>`

`<nav></nav>`

`<section></section>`

`<details></details>`



HTML Audio

The HTML `<audio>` element is used to play an audio file on a web page.

```
<audio controls>
```

```
<source src="horse.ogg" type="audio/ogg">
```

```
<source src="horse.mp3" type="audio/mpeg">
```

Your browser does not support the audio element.

```
</audio>
```

- The `controls` attribute adds audio controls, like play, pause, and volume.
- The `<source>` element allows you to specify alternative audio files which the browser may choose from. The browser will use the first recognized format.
- The text between the `<audio>` and `</audio>` tags will only be displayed in browsers that do not support the `<audio>` element.

HTML <audio> Autoplay

To start an audio file automatically, use the **autoplay** attribute:

```
<audio controls autoplay>
```

```
<source src="horse.ogg" type="audio/ogg">
```

```
<source src="horse.mp3" type="audio/mpeg">
```

Your browser does not support the audio element.

```
</audio>
```

Add **muted** after **autoplay** to let your audio file start playing automatically (but muted):

```
<audio controls autoplay muted>
```

```
<source src="horse.ogg" type="audio/ogg">
```

```
<source src="horse.mp3" type="audio/mpeg">
```

Your browser does not support the audio element.

```
</audio>
```

HTML Video

The HTML `<video>` element is used to show a video on a web page.

```
<video width="320" height="240" controls>  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.ogv" type="video/ogg">  
  Your browser does not support the video tag.  
</video>
```

- The `controls` attribute adds video controls, like play, pause, and volume.
- It is a good idea to always include `width` and `height` attributes. If height and width are not set, the page might flicker while the video loads.
- The `<source>` element allows you to specify alternative video files which the browser may choose from. The browser will use the first recognized format.
- The text between the `<video>` and `</video>` tags will only be displayed in browsers that do not support the `<video>` element.

HTML <video> Autoplay

To start a video automatically, use the **autoplay** attribute

```
<video width="320" height="240" controls autoplay>
```

```
<source src="movie.mp4" type="video/mp4">
```

```
<source src="movie.ogg" type="video/ogg">
```

Your browser does not support the video tag.

```
</video>
```

Remove **controls** to remove the player.

Add **muted** after **autoplay** to let your video start playing automatically (but muted)

```
<video width="320" height="240" autoplay muted>
```

```
<source src="movie.mp4" type="video/mp4">
```

```
<source src="movie.ogg" type="video/ogg">
```

Your browser does not support the video tag.

```
</video>
```


How embed Google Map

- Go to **google map**.
- Search the desired location.
- Click on **share** button on it.
- Click on **embed a map**.
- You can select proper size like medium, large, custom size etc. Preferably select custom size so that we can adjust it as per our requirement in HTML.
- Click on **Copy HTML** now paste this link in our html and to see the result refresh the webpage.

HTML <iframe> Tag

The **<iframe>** tag specifies an inline frame.

An inline frame is used to embed another document within the current HTML document.

Syntax: **<iframe src=" " > </iframe>**

Note we can include any website which is not restricted.

Eg:

```
<iframe src="https://www.ndtv.com" height="500" width="1000">  
</iframe>
```

Embed PDF

There are 3 ways to embed pdf in webpage

- Using **<embed>** tag
- Using **<iframe>** tag
- Using **<Object>** tag

Using **<embed>** tag

```
<embed src="test.pdf" type="application/pdf"
width="1000" height="500">
```

Note: No closing tag required.

Using **<iframe>** tag

```
<iframe src="test.pdf" height="800" width="800">
</iframe>
```

Note: No type is required as in case of embed.

Using **<Object>** tag

```
<object data="test.pdf" height="1200" width="500"
type="application/pdf"> </object>
```

Note: No src instead data attribute.

How Embed YouTube video

There are 2 types of videos on YouTube.

- One which you can embed.
- Another which you cannot embed.

Note: While uploading video on YouTube there is an option whether to **allow embed or not**. If user restrict embed then that video cannot be embedded.

Steps to embed:

- Open any video in YouTube.
- Click on **share**
- Click on **embed**
- A code (iframe) will appear copy it and paste it in your html.
- Save and refresh the browser to see the output.
- Optionally you can set the **start at** time option and other options.